

MUHAMMED SAHAL MM

+91 9544686536 | sahalmusawirofficial@gmail.com | linkedin.com/in/muhammedsahalmm | github.com/muhammedsahalmm

SUMMARY

Postgraduate in Computer Science specializing in AI, Machine Learning, and Data Science, with hands-on experience in building and deploying end-to-end machine learning models. Proficient in Python, Scikit-learn, NLP, and modern LLM tools (including RAG, LangChain, and Ollama). Skilled at solving real-world problems through data-driven insights and predictive analytics. Seeking an AI/ML Engineer or Data Scientist role to contribute impactful solutions.

TECHNICAL SKILLS

Languages: Python, MySQL, C, C++

Machine Learning / AI: Scikit-learn, Computer Vision, LangChain, NLTK, spaCy, Ollama, FAISS

Data Analysis / Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Tableau, Excel

Development / Deployment: Flask, FastAPI, Streamlit, Bootstrap, Git, GitHub, Tkinter, HTML, CSS

Platforms / Tools: Jupyter, Google Colab, Anaconda, Excel, Visual Studio, PyCharm, SQLyog, Streamlit Cloud

EXPERIENCE

TECHOLAS TECHNOLOGIES, Kozhikode

November 2025 - Present

Data Analytics Intern

- Working on real-world datasets to perform data cleaning, preprocessing, and exploratory data analysis (EDA) to support business decision-making.
- Creating data visualizations and dashboards using Python and analytical tools to identify trends, patterns, and insights.

IBM, Bengaluru

June 2025 - July 2025

AI ML Intern (Hybrid)

- Built and deployed an NLP & ML system as an interactive Streamlit app, enhancing accessibility and user engagement.
- Applied Python, Scikit-learn, Pandas, and GitHub workflow for end-to-end model development and deployment.

PROJECTS

Enterprise Coding Assistant with RAG (Local LLM)

FastAPI, Tkinter, LangChain, Ollama, FAISS, Nomic Embeddings

- Built a local coding assistant using RAG architecture, integrating LangChain, FAISS, Ollama, and DeepSeek for secure enterprise-level code generation, semantic search, Q&A, and document retrieval with a FastAPI backend and Tkinter GUI.
- Designed a custom knowledge base with Nomic Embeddings, enabled secure local LLM processing to eliminate data safety concerns, and packaged the solution as a standalone desktop app using PyInstaller, supporting both web and desktop interfaces for flexible deployment.

Book Genre Predictor

Python, Scikit-learn, MultinomialNB, NLP, Streamlit, Git, GitHub

- Built an NLP-based Streamlit app to predict book genres using TF-IDF and Multinomial Naive Bayes.
- Implemented preprocessing, model training, version control with Git/GitHub, and live deployment on Streamlit Cloud. Configured `.gitignore` to securely handle sensitive files.
- Live App: [bookgenrepredictor](#)

Customer Churn Prediction Model

HTML, CSS, Python, Flask, Google Colab, Visual Studio, Scikit-learn, RandomForest

- Built and deployed a Flask-based machine learning model using RandomForestClassifier and SMOTE to predict customer churn based on usage patterns.
- Achieved high model performance (accuracy, precision, recall, F1-score) and enabled real-time predictions to support proactive retention strategies.

Image Classification ML Model

HTML, CSS, Python, Flask, Google Colab, Visual Studio, Support Vector Machine

- Developed a Machine Learning model using image processing, feature extraction, and SVM to classify images as Aeroplane or Car.
- Created a Flask-based web app where users input an image URL, which is processed and classified instantly by the trained model.

Fever Medication Recommendation AI Model

Google Colab, Visual Studio, Logistic Regression

- Built an AI model using Logistic Regression to predict suitable fever medication based on age, BMI, symptoms, and chronic conditions.
- Performed Exploratory Data Analysis and selected key features to improve model accuracy and interpretability.

Event Management System

HTML, CSS, Bootstrap, Python, Flask, SQLyog

- Built a website using Python, featuring different roles such as Admin, Users, Public, Event Agencies, and Workers.
- The admin checks customer access features, allowing users and the public to easily access resources. Event agencies can add or remove their own resources.

EDUCATION

Yenepoya University Bangalore

Yenepoya University

Master of Computer Science | 2023 - 2025

Jamia Hamdard Kannur

Jamia Hamdard University

Bachelor of Computer Application | 2020 - 2023

GHSS Pallikunnu

Kerala Board

Higher Secondary in Computer Science | 2018 - 2020

CERTIFICATIONS & TRAINING

IBM Data Analyst Professional Certificate

January 2025

Verify at Coursera, IBM: [ibmdataanalyst](#)

- Completed 11 courses covering Excel, SQL, Python, and Cognos Analytics.
- Built dashboards and analyzed real-world datasets for business decision-making.

Data Science & AI, Tableau Workshop

November 2024

YENINTELLECT, Yenepoya University

- Hands-on Tableau workshop focused on AI and data-driven visualization.
- Built interactive dashboards and interpreted datasets to support decisions.

Python & Flask Training

June 2022 - Oct 2022

Riss Technologies, Kannur

- Built a role-based event management web app using Flask and Python.
- Gained backend experience handling user roles, permissions, and resource management.

ONLINE COURSES

- **Exploratory Data Analysis for Machine Learning** | Coursera: [verify](#)
- **Deep Learning and Re-inforcement Learning** | Coursera: [verify](#)
- **Fundamentals of Visualization with Tableau** | Coursera: [verify](#)
- **Introduction to Cloud Computing** | Coursera: [verify](#)